Apply filters to SQL queries

Project description

In this project, I used SQL queries to find security issues and unauthorized logins in a company database. I filtered the data to look for things like failed login attempts and suspicious access from outside the country. This helped me quickly find security risks and make sure only the right employees had access to our systems.

Retrieve after hours failed login attempts

```sql
SELECT *
FROM log_in_attempts
WHERE login_time > '18:00' AND success = FALSE;
```

This query will get all the records in which the login time was after 6pm and there was no success in logging in.

Retrieve login attempts on specific dates

```sql
SELECT *
FROM log_in_attempts
WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

This query will check if there is any suspicious activity that happened on 09-05-2022 or on 08-05-2022.

Retrieve login attempts outside of Mexico

```sql
SELECT *
FROM log_in_attempts
WHERE NOT country LIKE 'MEX%';
```

This query will check all suspicious login attempts that happened outside mexico since the team determined that no suspicious activity occurred in mexico. I used the '%' to indicate that there are more letters after x since mexico can be written full or as mex.

Retrieve employees in Marketing

```
SELECT *
FROM employees
WHERE department = 'Marketing' AND office LIKE 'East%';
```

This query will check all records in which the department is Marketing and all offices in the East.

Retrieve employees in Finance or Sales

```
SELECT *
FROM employees
WHERE department = 'Finance' OR department = 'Sales';
```

This query will display all the records in which the department is finance or sales to perform a security update on the machines.

Retrieve all employees not in IT

```
SELECT *
FROM employees
WHERE NOT department = 'Information Technology';
```

This query will display all records in all departments except the Information Technology department since this is the only department that had their machines updated.

Summary

I ran SQL commands using operators like `AND`, `OR`, `NOT`, and `LIKE` to filter through employee and login records. I successfully found late-night failed logins, logins from outside Mexico, and sorted employees by their departments to check who has access. These basic SQL skills are really helpful for quickly finding security problems and managing user permissions.